Fuzzy Concepts in Expert Systems

A Paper Review

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Summary

Fuzzy Concepts in Expert Systems

Authors: by K. S. Leung and W. Lan, both from the Chinese University of Hong Kong (CUHK)

Published: September 1988

Subject: An implementation of Fuzzy Logic to design an Expert System better equipped to deal with representations of inexact knowledge present in natural language.

Keywords

Fuzzy Logic, Artificial Intelligence, Systems engineering

A Brief Introduction to Expert Systems (ES)

> Source: Internet meme from Iron Men 2

Core components of ESs

Knowledge base

A collection of stored assertions, or `facts about the world`.

Inference engine

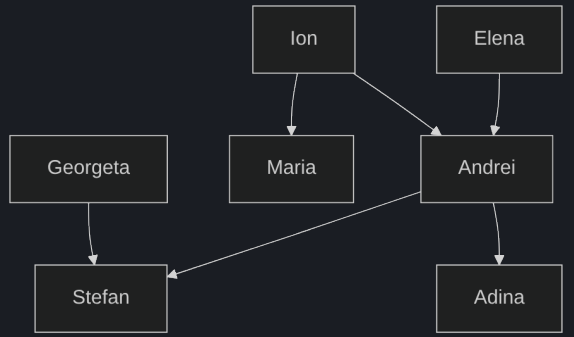
An Automated reasoning system to evaluate conclusions given the current state of the knowledge base and the recursive application of relevant `rules of inference`.

- Rules of inference are glorified `if/then` statements.
- Two main modes of operation: **forwards** and **backwards chaining**.

> Leung and Lan's solution make use of **backwards chaining**.

An example

Family tree diagram



Prolog code

```
% --- Facts (The Database) ---
% parent(Parent, Child)
parent(ion, andrei).
parent(elena, andrei).
parent(ion, maria).

parent(andrei, stefan).
parent(andrei, adina).
parent(georgeta, stefan).

% --- Rules (The Logic) ---
% A is a grandparent of B if A is a parent of C,
and C is a parent of B.
grandparent(GP, GC) :-
    parent(GP, Parent),
    parent(Parent, GC).

:- initialization(main).
```

An example

Querying

```
main :-
    format('~n--- Query 1: grandparent(X,
stefan) ---~n'),
    grandparent(X, stefan),
    format('X = ~w~n', [X]),
    fail.
main :-
    format('~n--- Query 2: grandparent(ion, X)
---~n'),
    grandparent(ion, X),
    format('X = ~w~n', [X]),
    fail.
main :-
    format('~n--- Query 3: grandparent(GP, GC)
---~n'),
    grandparent(GP, GC),
    format('GP = ~w, GC = ~w~n', [GP, GC]),
    fail.
main :-
    halt.
```

Results

```
Execution time: seg 13 abr 2026 14:24:36 EEST

--- Query 1: grandparent(X, stefan) ---
X = ion
X = elena

--- Query 2: grandparent(ion, X) ---
X = stefan
X = adina

--- Query 3: grandparent(GP, GC) ---
GP = ion, GC = stefan
GP = ion, GC = adina
GP = elena, GC = stefan
GP = elena, GC = adina
```

Accessory components of ESs

Not strictly necessary, but *very much required* in real world applications:

- Knowledge Acquisition and Truth Maintenance utilities,
- User Interface and Hypothetical Reasoning utilities,
- Ontology classification, and *finally*,
- **Uncertainty Systems** (i.e.: the crux of System Z-II)

Heuristics and inexact knowledge

Much of human knowledge is **inexact** and, therefore, **cannot be represented with first-order logic statements.**

Heuristics

Approximate solutions operating on incomplete or imperfect information.

For that the authors identify two sources of inexact knowledge:

Uncertainty

The system is unsure about a given predicate. Example using a Confidence Factor (CF):

```
rule:
    IF X is a bird,
    THEN it can fly. (CF = 0.9)

fact:
    X is a bird. (CF = 0.8)

conclusion:
    X can fly. (CF = 0.9 × 0.8 = 0.72)
```

Vagueness (fuzzy concepts)

The system is unsure about the boundaries of a given predicate. Example:

```
rule:
    IF
        (body is well-built OR height is tall)
        AND person is healthy
    THEN
        weight is heavy
```

Vagueness is subject to the Soristes Paradox (a.k.a. the *Heap Paradox*)

Interactions between uncertainty and vagueness

Previous ESs, such as Cadiag-2, had some capability to deal with fuzzy concepts. System Z-II excels at it by also being able to deal with:

Uncertainty about vagueness or vagueness about uncertainty

```
rule:
    IF
        (body is well-built OR height is tall)
        AND person is healthy
    THEN
        weight is heavy
    WITH
        CERTAINTY → close to 0.8
```

Vagueness about vagueness

Statements such as "John is very tall" or "John is quite tall".

Approaches to uncertainty

Conditional probability

Methods such as the Bayesian Inference and the Dempster-Shafer Theory of Evidence.

- Empirical methods, often associated with Machine Learning Algorithms
- Require massive amounts of parameterized data in order to be effective.

Confidence Factors

Assigning certainty based on the opinions of experts.

> This was the approach taken by System Z-II

System Z-II's approach to vagueness

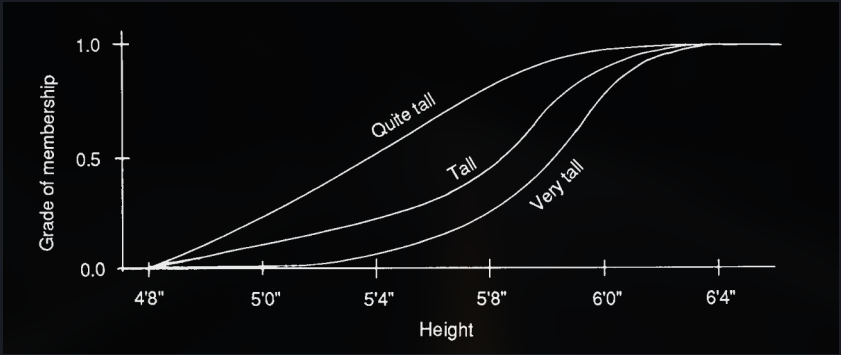
Fuzzy sets with adverbs as arithmetic modifiers

Take the following example for the fuzzy concept `tall` represented as a set with different degrees of membership ranging from 0.0 to 1.0.

Tall membership values for different heights and use of adverbs

Height	No adverb	quite (square root)	very (squared)
⩽ 4'8"	0.0	0.0	0.0
5'0"	0.1	0.31	0.01
5'4"	0.2	0.44	0.04
5'8"	0.7	0.83	0.49
6'0"	0.9	0.94	0.81
⩾ 6'4"	1.0	1.0	1.0

Tall membership values for different heights and use of adverbs



Source: from the paper

Relationships between fuzzy data and non-fuzzy consequents

Consider the following rule and fact:

```
rule:
    IF X is tall,
    THEN X should be chosen as a member of the basketball team.
    (CF_1 = 1.0)
fact:
    Stefan is quite tall
    (CF_2 = 1.0)
conclusion:
    Stefan should be chosen as a member of the basketball team.
    (CF_3 = y)
```

> Which value should y have?

Similarity

For Leung and Lan, the CF for the conclusion should be the product between the rule and fact CFs and a Similarity (M) factor:

$$CF_3 = CF_1 \cdot CF_2 \cdot M$$

In turn, Similarity is calculated using the following algorithm:

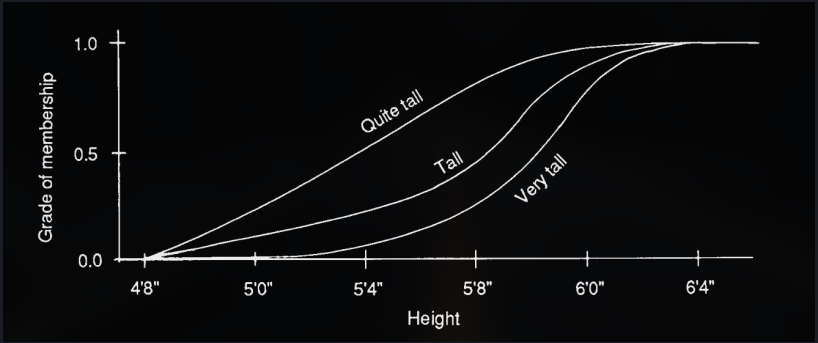
```
IF N(F|F') > 0.5
THEN M = P(F|F')
ELSE M = (N(F|F') + 0.5) * P(F|F')
```

Where N is the Necessity and P is the Possibility, for the fuzzy sets F and F' being matched.

Possibility

The maximum degree of intersection between two sets, expressed as

$$P(F \mid F') = \max_{w}(\min(\mu_F(w), \mu_{F'}(w)))$$



> The "quite tall" meets the "tall" at 6'4" at its peak and, therefore, has a possibility of 1.0

Necessity

The degree in which the data meets the pattern exactly, expressed as

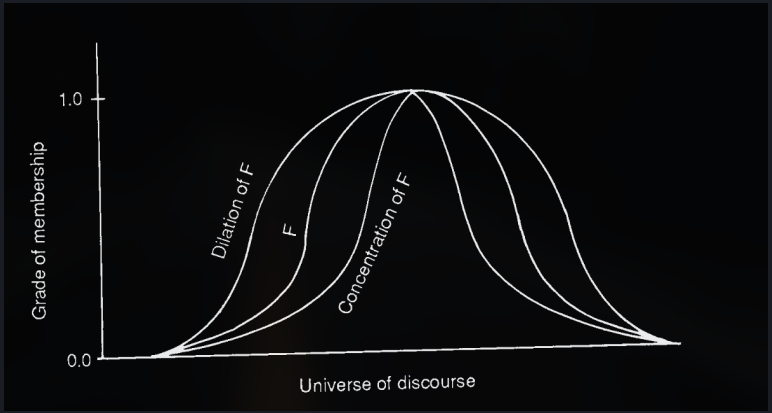
$$N(F \mid F') = 1 - P(\overline{F} \mid F')$$

Where $\overline{F}$ is the complement of F.

`N(F|F') > 0.5` → F' is a concentration of F.

`N(F|F') = 0.5` → F' is a duplicate of F.

`N(F|F') < 0.5` → F' is a dilation of F.



Source: original paper

Similarity

So, the confidence with which System Z-II can assert that Stefan should be chosen as a member of the basketball team is:

$P = 1.0$

$N = 1 - P(F \dagger F') = 0.3$

$N < 0.5 \rightarrow M = (N(F \dagger F') + 0.5) \times P(F|F') = (0.3 + 0.5) \times 1.0 = 0.8$ ■

Relationships between fuzzy data and fuzzy consequents

Consider this simple rule:

IF price is high, THEN profit is good

Such a rule requires the user to describe a couple of fuzzy sets and a rule relating those.

Knowledge acquisition

During the knowledge acquisition phase, the user can create the fuzzy sets and have them be related automatically using one of the three following rules (R_SG is the default):

$$R_{SG} = (F_1 \times V \underset{S}{\rightarrow} U \times F_2) \wedge (\overline{F} \times V \underset{G}{\rightarrow} U \times \overline{F}_2)$$

$$R_S = F_1 \times V \underset{S}{\rightarrow} U \times F_2$$

$$R_G = F_1 \times V \underset{G}{\rightarrow} U \times F_2$$

Where:

- `V` and `U` are the Universes of Discourse for `F_1` and `F_2` respectively.
- `×` is the Cartesian Product operator.
- `→_S` and `→_G` are the Standard and Gödelian variations of the Fuzzy Inference, respectively.

Consultation (inference)

During consultation, the conclusion C to the aforementioned rule can be expressed as their fuzzy composition:

$$\mu_C(x) = \max_w(\min(\mu_{F_1}(w)\mu_{F_2}(w, x)))$$

When applied, such a rule should map high price to good profit.

Calculating fuzzy confidence

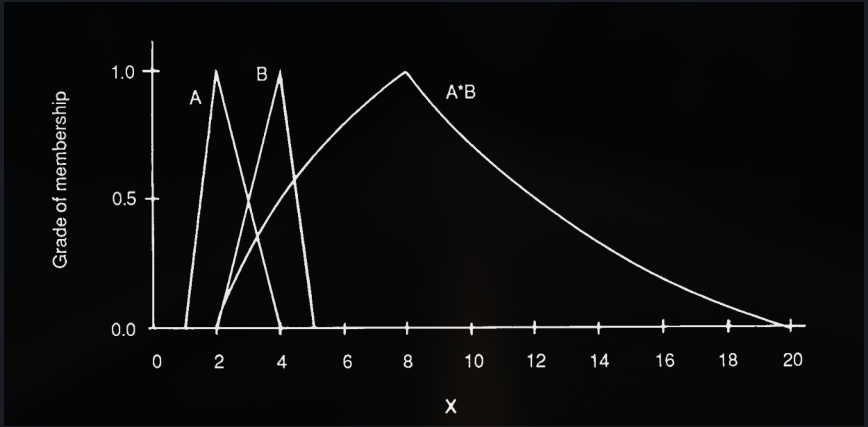
Consider the existence of a given rule and a given fact such that:

```
rule: IF A is V_1 THEN C is V_2 (FN_1)
fact: A is V_1                  (FN_2)
conclusion: C is V_2            (FN_3)
```

Let `(FN_X)` be fuzzy confidence values such as "close to 1.0".

Then the fuzzy confidence in the conclusion `C`, described by `FN_3`, is obtained though the process of `fuzzy number multiplication`:

$$\mu_{F_3}(z) = \max_{z=x\times y}\left(\min(\mu_{F_1}(x), \mu_{F_2}(y))\right)$$



Fuzzy number multiplication. source: original paper

As a result, "close to 2" and "close to 4" would map to "around 8". As 8 is the center of gravity, and "around" an adverb used to express the larger value spread.

Rules with multiple fuzzy propositions

```
rule:
    IF
        (body is well-built OR height is tall)
        AND person is healthy
    THEN
        weight is heavy
    WITH
        CERTAINTY → close to 0.8
```

AND → Fuzzy Union

$$\mu_U(x) = \min(\mu_{F_1}(x), \mu_{F2}(x))$$

OR → Fuzzy Intersection

$$\mu_I(x) = \max(\mu_{F_1}(x), \mu_{F2}(x))$$

> Fuzzy intersection is also used when multiple fuzzy rules corroborate with the same conclusion.

System Z-II's architecture

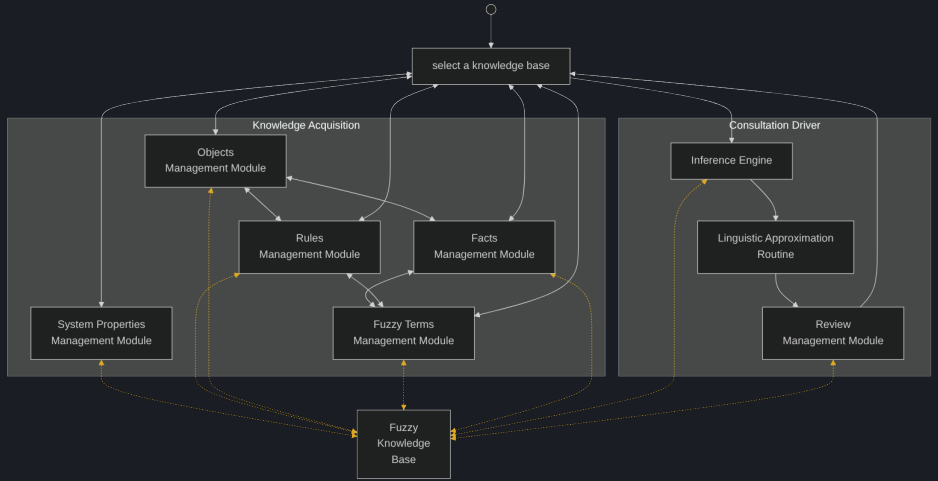
Source: Computer Architecture Webpage at Keio University

System Z-II was designed to operate in a **VAX-11/708** microcomputer.

At its *most basic description* is composed by three interacting subsystems:

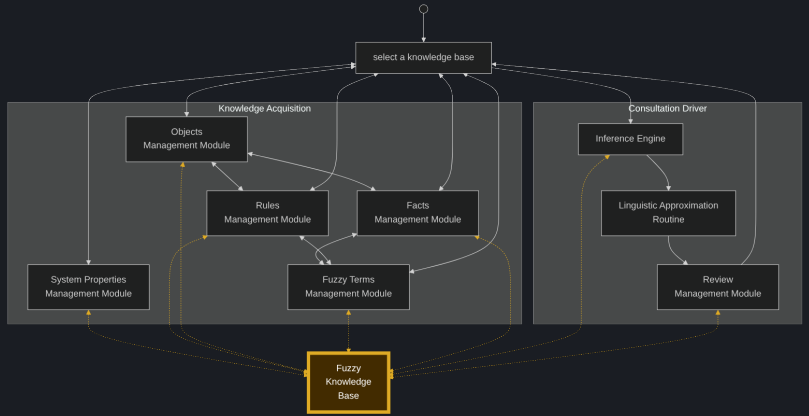
1. Knowledge acquisition
2. Consultation driver
3. Fuzzy knowledge base

Modules overview



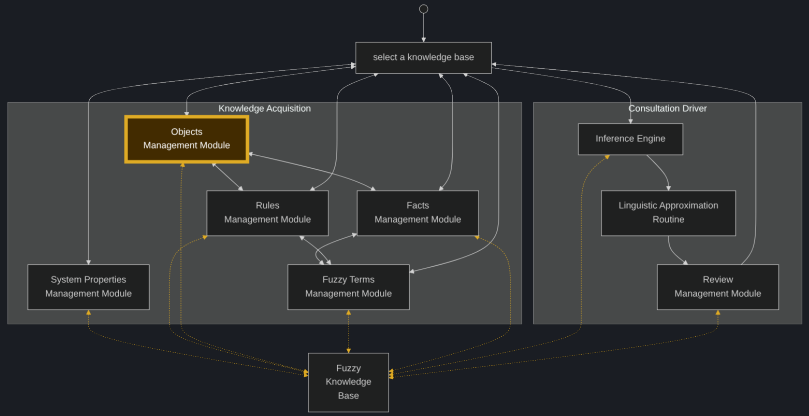
Legend: - - → : Data flow | ⟶ : Control flow

Fuzzy Knowledge Base



Stores the fuzzy terms and sets as hash tables and property lists.

Objects Management Module



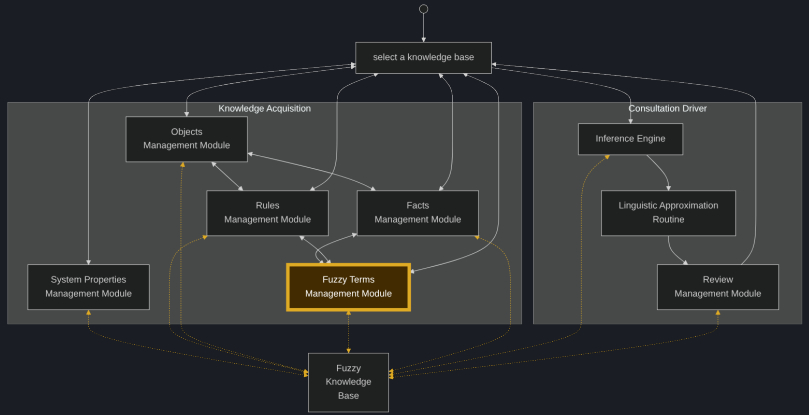
Creates, modifies and deletes objects in the system. Objects are named data structures endowed with attributes.

> Think in terms of Object Oriented Programming (OOP).

Predefined attributes

Attribute	Type	Contents
TYPE	String	Sets the object's type
FUZZY-OR-NOT	Boolean	Indicates if values stored are fuzzy
ASK-FIRST-OR-NOT	Boolean	Indicates if the user should be asked the object's values
USED-BY-RULES	[Rules]	Rules that contain this object as an antecedent
UPDATED-BY-RULES	[Rules]	Rules that contain this object as an consequent

Fuzzy Terms Management Module

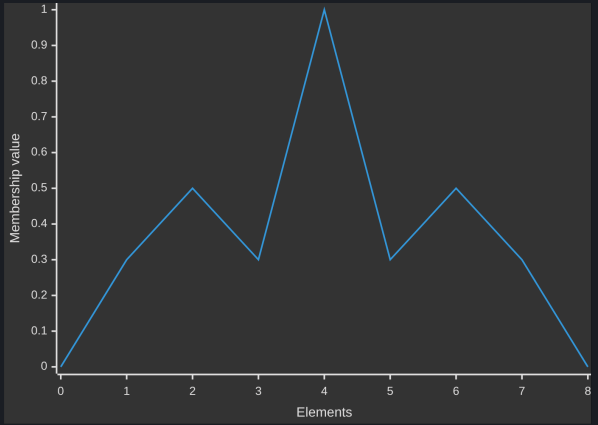


Creates and maps the existing fuzzy sets, associated with the fuzzy terms (i.e. "very tall", "rather good") seen in objects where FUZZY-OR-NOT is True.

Fuzzy Sets

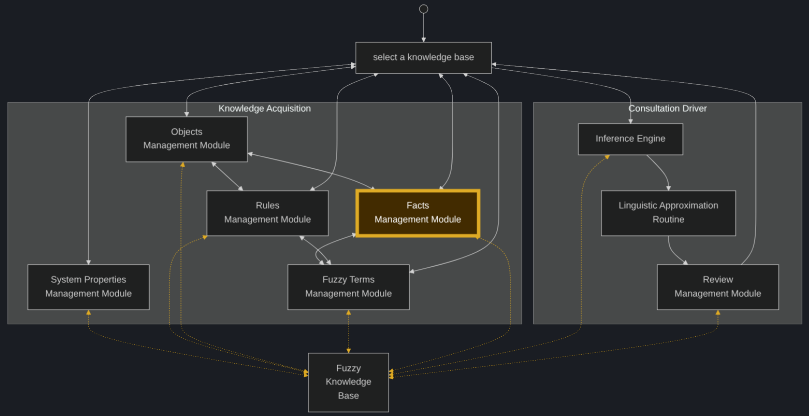
Fuzzy sets are implemented as lists of numbers that satisfy the conditions of normality and convexity.

- A fuzzy set A is said to be normal when *at least one of its elements* x has a membership grade of 1.0.
- A fuzzy set A is said to be convex when, for all values x, the membership function μ(x) does not "dip and then rise again".



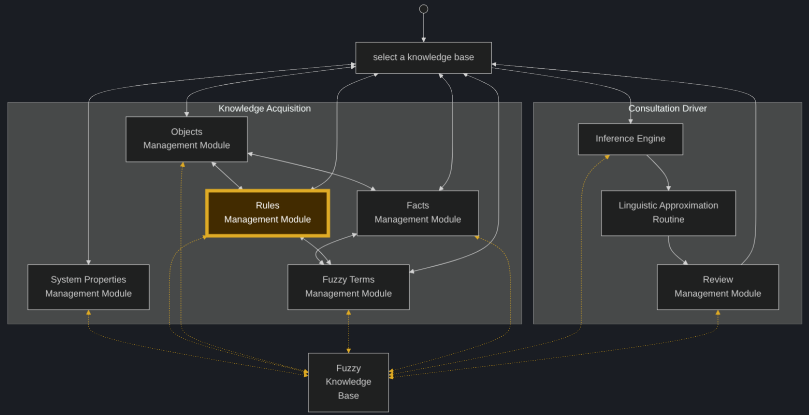
Non-convex fuzzy set example

Facts Management Module



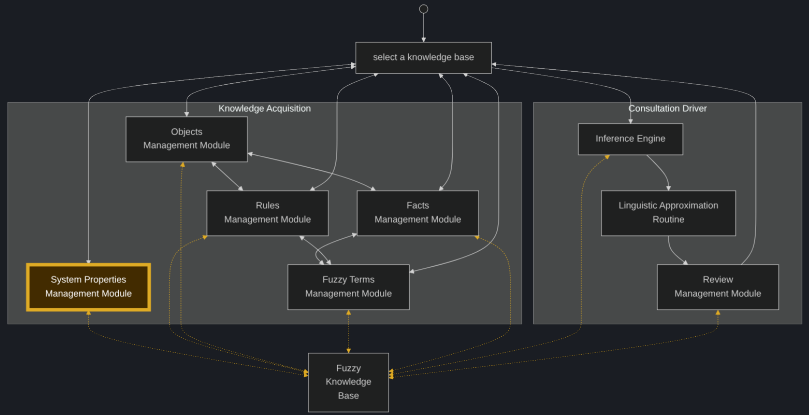
Creates and manages facts (assertions). Facts are established by the user through the use of restricted English sentences, either of his own initiative or when prompted by the system when more information is required.

Rules Management Module



Creates and manages facts (assertions). Facts are established by the user through the use of restricted English sentences, either of his own initiative or by being prompted by the system in a step-by-step manner when such information is required.

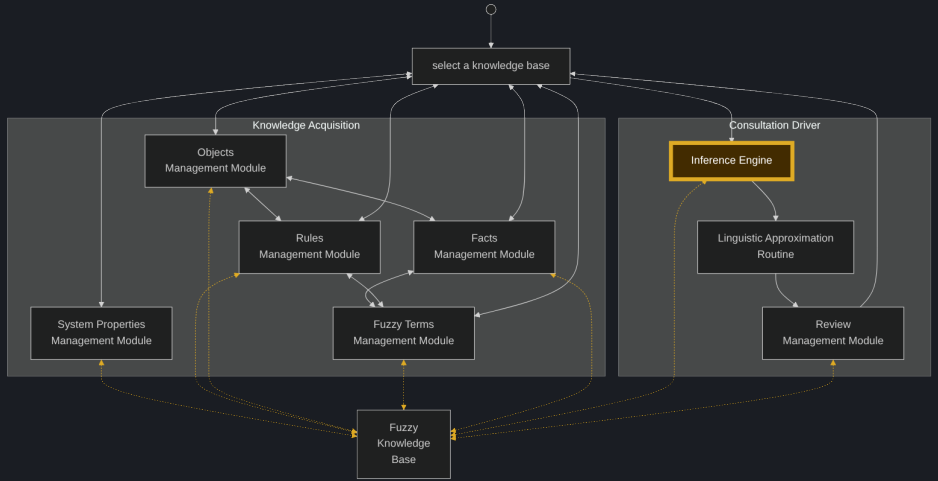
System Properties Management Module



Sets global settings such as:

- GOAL-OBJECTS: Objects associated with conclusions.
- INITIAL-ASK-OBJECTS: Objects associated with propositions.
- DOMAIN-DESCRIPTION: Descriptions of the domains of knowledge covered by the system

The Inference Engine



Reasoning chain

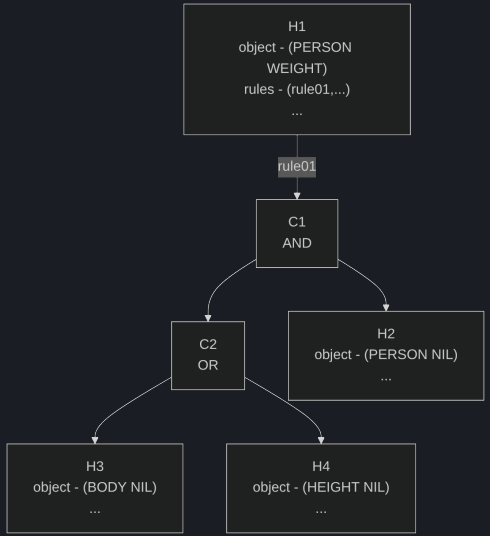
- **Backwards chaining:** questioning is guaranteed to follow a focused goal conclusion.
- Goals might be set by the user or automatically fetched from the SPM module.
- An example of user query would be: "What should the <ATTRIBUTE> of <OBJECT> be?", i.e. "What should the weight of the person be?"
- The system then constructs a reasoning chain from the conclusion to its propositions following the pointers set by the default `UPDATED-BY_RULES` attribute.

Query

What should the weight of the person be?

Rule 1

```
rule01:
    IF
        (body is well-built OR height is tall)
        AND person is healthy
    THEN
        weight is heavy
```

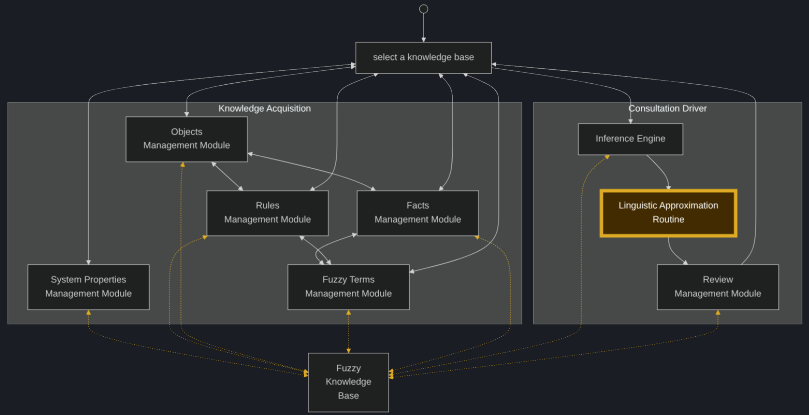


Rule evaluation

After the reasoning tree is constructed the result is constructed through `forward evaluation`, where:

- The user is asked questions that take from the various possible propositions toward the conclusion.
- All the relevant Fuzzy calculations are performed.

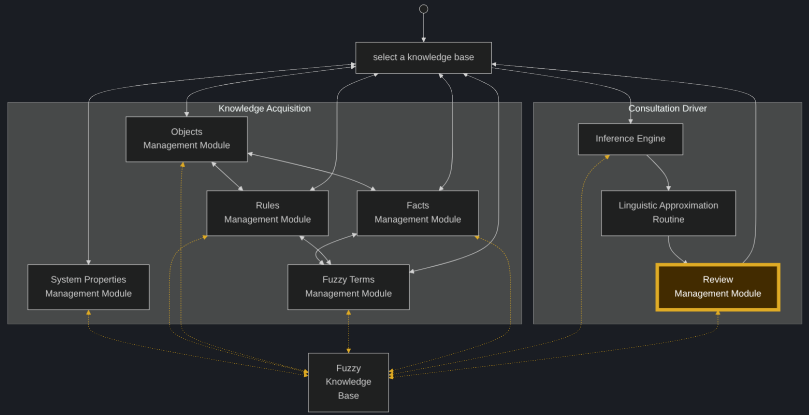
Linguistic Approximation Routine



Maps the fuzzy sets and fuzzy numbers produced during the inference phase into linguistic descriptions.

This is done on the basis of the location of the fuzzy set's center of gravity (maps to an adjective or number) and its imprecision (the sum of the membership values — maps to an adverb)

Review Management Module



Allows the user to trace reasoning chains, consult *why* and *how* facts where established, and alter those to handle *what-if* scenarios.

Working example

The authors provide a script to setup System Z-II as a career advisor. It contains about 60 rules, most of them using fuzzy terms and some using fuzzy numbers as well.

Its expected output is, by the end of the consultation process, produce a list of career choices listed from the most to the least likely, like so:

```
0.97 certain: Department of Physics,
0.95 certain: Department of Computer Science,
0.93 certain: Department of Mathematics,
0.90 certain: Department of Electronics
0.86 certain: Department of Biology
0.72 certain: Department of Chemistry
Very close to 0.70 certain: Faculty of Arts
Very close to 0.69 certain: Faculty of Social Science
Close to 0.49 certain: Faculty of Business
0.48 certain: Faculty of Medicine
```

Full disclosure

As I did not have my own VAX-11/780, or knowledge of a git repository with System Z-II's code, or the time to reimplement the whole thing in Prolog for the sake of doing so, this should mark the end of our presentation.

Vă mulțumesc pentru timpul acordat.